

OS PRACTICAL

PRACTICAL-6

6. Create a client-server application using Named Pipes (FIFOs). The client sends messages and the server processes and responds to them. Analyze FIFO behavior when multiple clients communicate simultaneously.

Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

```
rishitha@Rishitha:~$ cd ~
rishitha@Rishitha:~$ mkdir practical5
mkdir: practical5: File exists
rishitha@Rishitha:~$ mkdir practical6
rishitha@Rishitha:~$ cd practical6
rishitha@Rishitha:~/practical6$ pwd
/home/rishitha/practical6
rishitha@Rishitha:~/practical6$ nano server.c
rishitha@Rishitha:~/practical6$ nano client.c
rishitha@Rishitha:~/practical6$ gcc server.c -o server
gcc client.c -o client
server.c: In function 'main':
server.c:43:52: warning: '%s' directive output may be truncated writing up to
 43 |         snprintf(response, SIZE, "Server received: %s", message);
      |                                         ^~          ~~~~~
server.c:43:9: note: 'snprintf' output between 18 and 117 bytes into a desti
 43 |         snprintf(response, SIZE, "Server received: %s", message);
      |         ^~~~~~
rishitha@Rishitha:~/practical6$ gcc server.c -o
gcc: error: missing filename after '-o'
rishitha@Rishitha:~/practical6$ gcc server.c -o server
server.c: In function 'main':
server.c:43:52: warning: '%s' directive output may be truncated writing up to
 43 |         snprintf(response, SIZE, "Server received: %s", message);
      |                                         ^~          ~~~~~
server.c:43:9: note: 'snprintf' output between 18 and 117 bytes into a desti
 43 |         snprintf(response, SIZE, "Server received: %s", message);
      |         ^~~~~~
rishitha@Rishitha:~/practical6$ nano server.c
rishitha@Rishitha:~/practical6$ gcc server.c -o server
rishitha@Rishitha:~/practical6$ ls
client  client.c  server  server.c
rishitha@Rishitha:~/practical6$ ./server
Server started...
Client message: Hello Server
Client message: Hello from Client 2
|
```

```
rishitha@Rishitha:~$ cd ~/practical6
rishitha@Rishitha:~/practical6$ ./client
Enter message: Hello Server
Server response: Server received: Hello Server
rishitha@Rishitha:~/practical6$
```

```
rishitha@Rishitha:~$ cd ~/practical6
rishitha@Rishitha:~/practical6$ ./client
Enter message: Hello from Client 2
Server response: Server received: Hello from Client 2
rishitha@Rishitha:~/practical6$ █
```

[illegible]

```
rishitha@Rishitha:~$ cd ~/practical6
rishitha@Rishitha:~/practical6$ kill -SIGUSR1 1002
rishitha@Rishitha:~/practical6$ kill -SIGTERM 1002
rishitha@Rishitha:~/practical6$ kill -9 1002
rishitha@Rishitha:~/practical6$
```

```
Program is running...  
  
SIGUSR1 received: User-defined signal  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
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Program is running...
```

```
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
  
SIGTERM received: Termination signal  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
^C  
SIGINT received: Interrupt signal  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Killed ./signals  
rishitha@Rishitha:~/practical6$
```